Exam Date & Time: 03-Jan-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (CORE STREAM) DEGREE EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: BIO 1071- BIOLOGY FOR ENGINEERS

BIOLOGY FOR ENGINEERS [BIO 1071]

Marks: 50 **Duration: 180 mins.**

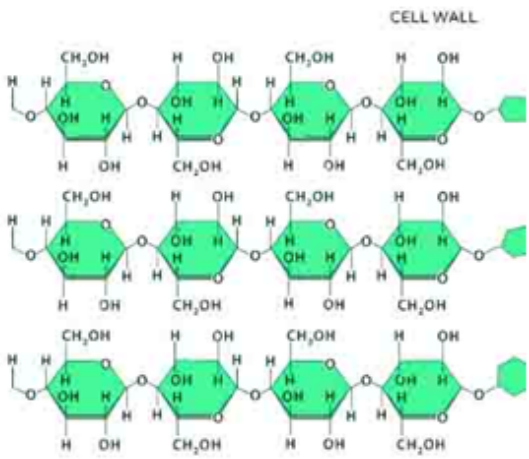
A

Answer all the questions.

1A) Discuss three major reasons for carbon as the major element of life on earth (3)

1B) Some proteins attains alpha helix with hydrogen bonding,but some others results in beta pleated sheet. How? (3)

1C) Given below is an illustration representing the arrangement of three layers of cellulose nanofibrils (4)



i) How the hydrogen bonding happens here? Copy the figure and illustrate
ii) Even though it has polar groups, cellulose is insoluble. How?

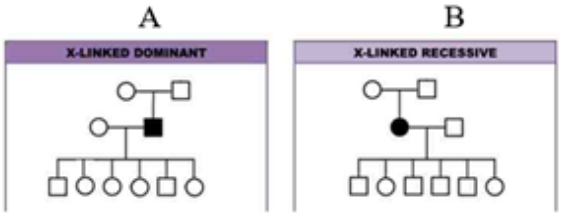
2A) Given below are the seeds produced from Mendel's pea plant on crossing between two individuals (3)

A B

Character : Seed shape
i) Write the genotype of the parents
ii) Illustrate the cross
iii) Write the genotype of the above seed

2B) Compare and contrast autosomal and X linked mode of inheritance (3)

2C) Given below are two pedigree charts A and B, but some individuals are not shaded.



(4)

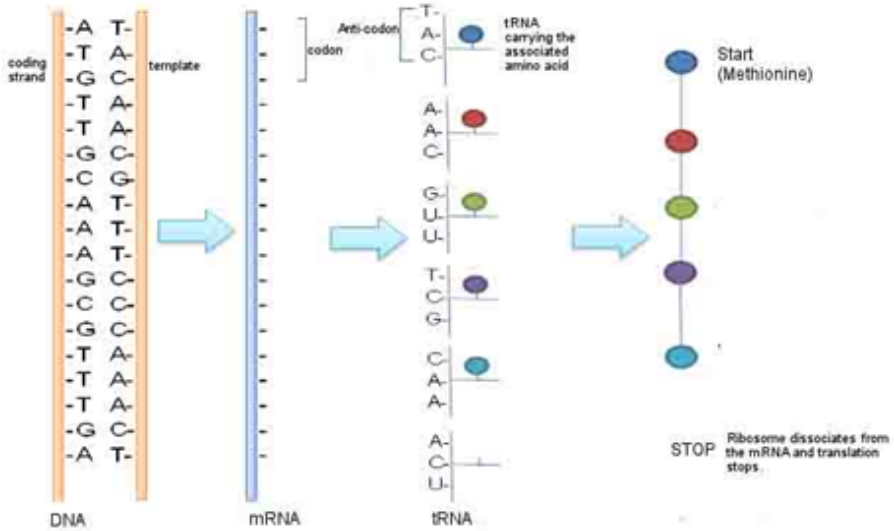
i) Complete the chart appropriately
ii) Give a valid justification in each case
(2+2 = 4 marks)

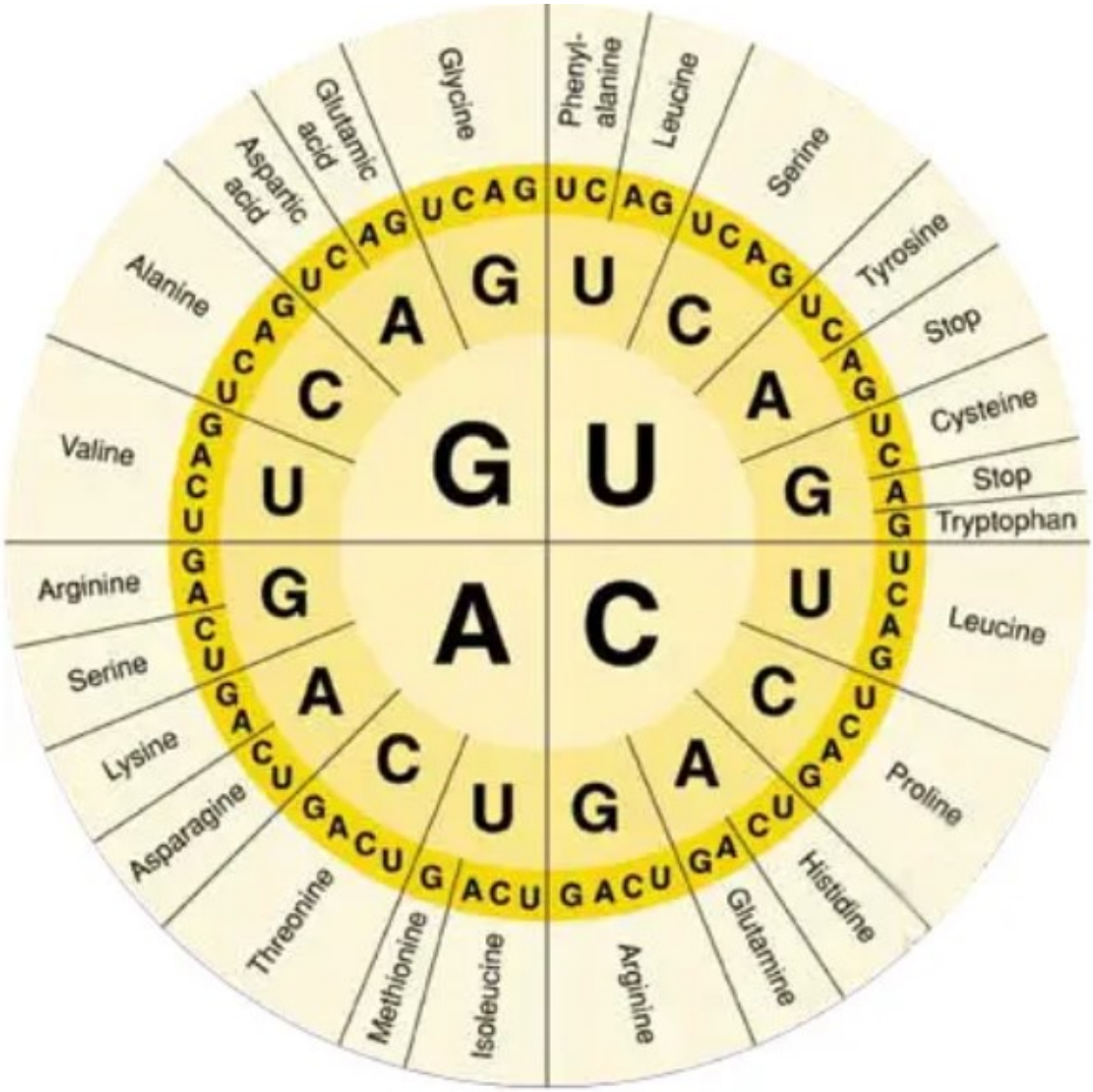
3A) Analyze the following replication fork and label the leading and lagging strands, RNA primers, location of helicase and okazaki fragments

(3)

3B) Mature mRNA is always shorter than newly formed mRNA. Illustrate with explanation (3)

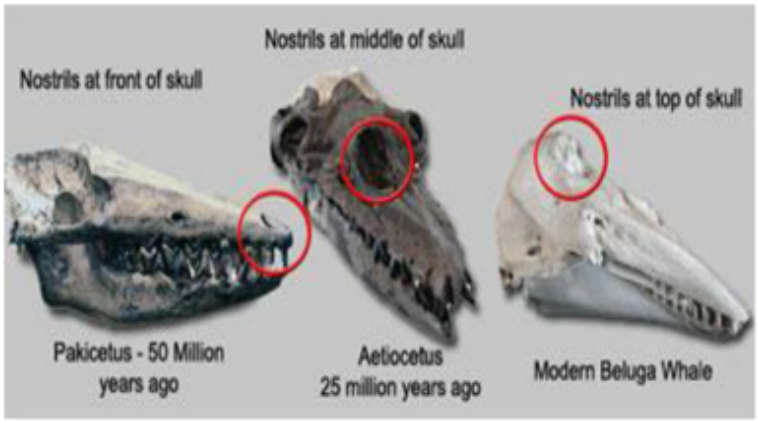
3C) Given below is an illustration of protein synthesis (4)
Copy the figure and illustrate the following
i) Mature mRNA sequence
ii) Protein sequence (First four amino acids after start)
iii) Label the ends of protein
iv) Label the ends of anticodons





4A) Logically explain the inheritance of sickle cell anemia and malarial resistance (3)

4B) Several evidence are there for evolution. According to the below mentioned picture, Which type of evolution it specifies and what is the concept behind this? (3)



4C) Critically evaluvate mutations as a tool for improvising the existing life forms (4)

5A) i) When a suspension of microorganisms that induces antibody production to protect against disease, what does it leads to?
ii) Can killed microorganisms induce a disease while using for vaccination? Justify?
iii) Some vaccines are 100% effective, while others not. Justify
(1+1+1 = 3 marks) (3)

5B) Explain the biomimetic effect of the given pictures with example.

i.

ii.

(3)

(1½ x 2 = 3 marks)

5C) Give four examples for the inventions inspired from living organisms. Explain (4)

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